

Mosaic Backend

Complete Project Documentation

A Django REST Framework-based POS and Inventory Management System for Indian GST Compliance

Document Information

Generated:	April 28, 2026
Project Path:	C:\Users\Paper\StudioProjects\mosaic_backend
Platform:	Windows (win32)
Database:	SQLite3 (db.sqlite3)

1. Project Overview

Mosaic Backend is a comprehensive **Django REST Framework-based POS (Point of Sale) and Inventory Management System** specifically designed for **Indian GST (Goods and Services Tax) compliance**. This production-ready backend API supports inventory management with batch tracking, variants, and serial numbers; POS operations with cart/hold functionality and shift management; sales invoicing with automatic GST calculation; purchase management with PO, GRN, and Purchase Invoice workflow; multi-location support with different GSTINs per warehouse; and party management for customers and suppliers with credit tracking.

2. Technology Stack

Component	Technology	Version
Language	Python	3.14.4
Web Framework	Django	5.0+
API Framework	Django REST Framework	3.14+
Database	SQLite3	(default)
Image Processing	Pillow	12.0+
Filtering	django-filter	24.0+

3. Directory Structure

mosaic_backend/ - Main Django project directory

- settings.py - Django settings (main config)
- urls.py - Root URL routing
- wsgi.py - WSGI entry point

- asgi.py - ASGI entry point

stock/ - Stock/Inventory app

- models.py - 14 models (Category, Unit, Item, etc.)
- views.py - DRF ViewSets
- serializers.py - DRF Serializers
- urls.py - API routes
- authentication.py - Bearer token auth
- management/commands/ - create_api_config.py

sale/ - Sales/POS app

- models.py - 20+ models
- views.py - DRF ViewSets
- serializers.py - DRF Serializers
- urls.py - API routes (40+ endpoints)
- management/commands/ - seed_states.py

pos/ - POS Shift Management app

- models.py - Shift, CashTransaction
- views.py - ViewSets

configuration/ - Configuration app

- models.py - Warehouse, ApiConfiguration
- views.py, serializers.py, urls.py

master/ - Master data (shared)

- urls.py - Routes to shared endpoints

Other Files:

- manage.py - Django management script
- requirements.txt - Dependencies
- db.sqlite3 - SQLite database
- .gitignore - Git ignore rules

4. Installed Apps

App	Purpose
django.contrib.admin	Django admin interface
django.contrib.auth	Authentication system
django.contrib.contenttypes	Content type framework
django.contrib.sessions	Session management
django.contrib.messages	Messaging framework
django.contrib.staticfiles	Static file serving
rest_framework	Django REST Framework
django_filters	Advanced filtering
stock	Inventory/Stock management
sale	Sales/POS operations
master	Master data (shared)
configuration	Configuration management
pos	POS Shift management

5. Key Models

5.1 Stock App Models (14 models)

Model	Purpose
Category	Item categorization
Unit	Measurement units (pcs, kg, liters) with conversion
AttributeType	Variant types (Color, Size)
AttributeValue	Variant values (Red, Large)
TaxCode	HSN/SAC codes
TaxComponent	CGST/SGST/IGST rates
Item	Core product
ItemVariant	Product variants
ItemVariantAttribute	Variant attributes
Batch	Inventory batches (FIFO/LIFO)
ItemImage	Product images with WebP
OpeningStock	Initial stock entries
StockMovement	Stock transaction history
SerialNumber	IMEI/Serial tracking

5.2 Sale App Models (20+ models)

Model	Purpose
State	Indian states
BusinessLocation	Multi-GSTIN locations
Party	Customer/Supplier
Order	POS Cart/Hold
OrderItem	Cart items
Invoice	Sales invoice
InvoiceItem	Invoice line items
CreditNote	Sales return
Receipt	Payment received
PurchaseOrder	PO to supplier
GoodReceiptNote	GRN
PurchaseInvoice	Purchase invoice
DebitNote	Purchase return
PaymentOut	Payment to supplier

6. API Endpoints

6.1 Stock API (/v1/api/)

- **/v1/api/categories/** - GET/POST (list/create categories)
- **/v1/api/units/** - GET/POST (measurement units)
- **/v1/api/attribute-types/** - GET/POST (variant types)
- **/v1/api/tax-codes/** - GET/POST (HSN/SAC codes)
- **/v1/api/items/** - GET/POST (items/products)
- **/v1/api/items/{id}/variants/** - Nested variants
- **/v1/api/items/{id}/images/** - Nested images
- **/v1/api/batches/** - GET/POST (inventory batches)
- **/v1/api/opening-stock/** - GET/POST (initial stock)
- **/v1/api/stock-movements/** - GET/POST (transaction history)
- **/v1/api/serial-numbers/** - GET/POST (IMEI/serial tracking)

6.2 Sale API (/v1/sale/)

Parties:

- **/v1/sale/states/** - Indian states for GST
- **/v1/sale/business-locations/** - Multi-GSTIN locations
- **/v1/sale/parties/** - Customers/Suppliers
- **/v1/sale/parties/{id}/ledger/** - Party account statement

Sales/POS:

- **/v1/sale/orders/** - POS Cart/Hold
- **/v1/sale/orders/{id}/hold/** - Hold order
- **/v1/sale/orders/{id}/convert/** - Convert to invoice
- **/v1/sale/invoices/** - Sales invoices
- **/v1/sale/invoices/{id}/finalize/** - Finalize (deduct stock)
- **/v1/sale/invoices/{id}/cancel/** - Cancel invoice
- **/v1/sale/receipts/** - Payments received
- **/v1/sale/credit-notes/** - Sales returns

Purchasing:

- **/v1/sale/purchase-orders/** - Purchase orders
- **/v1/sale/grns/** - Good receipt notes
- **/v1/sale/purchase-invoices/** - Purchase invoices
- **/v1/sale/debit-notes/** - Purchase returns
- **/v1/sale/payments-out/** - Payments to suppliers

Reports:

- **/v1/sale/reports/daily-sales/** - Daily sales
- **/v1/sale/reports/gst-register/** - GST register
- **/v1/sale/reports/gstr1/** - GSTR-1 export

6.3 Configuration API (/v1/configuration/)

- **/v1/configuration/warehouses/** - Warehouse/Location management
- **/v1/configuration/warehouses/{id}/set-default/** - Set default
- **/v1/configuration/api-config/** - API token config

6.4 POS API (/v1/pos/)

- **/v1/pos/shifts/** - Shift management
- **/v1/pos/shifts/{id}/close/** - Close shift
- **/v1/pos/shifts/{id}/transactions/** - Cash transactions
- **/v1/pos/transactions/** - All cash transactions

7. Authentication & Security

Authentication: Bearer Token authentication is used to protect all `/v1/*/` endpoints. The `ApiKeyAuthentication` class extracts the Bearer token from the Authorization header and validates it against the `ApiConfiguration` singleton stored in the database.

Security Configuration:

- `DEBUG = True` (development mode)
- `ALLOWED_HOSTS = []` (accepts all hosts)
- API token stored in `ApiConfiguration` model (`pk=1`)

Generate/Regenerate Token:

```
python manage.py create_api_config --regenerate
```

8. Business Logic Features

GST Calculation:

- Intra-state (same state): CGST + SGST (each = half of total rate)
- Inter-state (different state): IGST (full rate)

Invoice Numbering:

- Format: `{GSTIN}/{Financial Year}/{Sequence}`
- Example: `27AAAAA0000A1Z5/2025-26/00001`
- Per-warehouse sequence tracking

Stock Tracking:

- Items WITHOUT variants: `current_stock` in `Item` model
- Items WITH variants: `current_stock` in `ItemVariant` model
- Batch tracking for FIFO/LIFO costing

Image Processing:

- Automatic WebP conversion on save (85% quality)
- Supports PNG, JPEG, and other formats

9. Management Commands

- `python manage.py create_api_config` - Create/regenerate API token
- `python manage.py seed_states` - Seed Indian states (36 states)
- `python manage.py migrate` - Run database migrations
- `python manage.py runserver` - Start development server
- `python manage.py createsuperuser` - Create admin user

10. REST Framework Configuration

Authentication Classes:

- `stock.authentication.ApiKeyAuthentication`

Permission Classes:

- `stock.authentication.ApiKeyPermission`

Pagination:

- Default: PageNumberPagination
- Page Size: 10

Filter Backends:

- DjangoFilterBackend
- SearchFilter
- OrderingFilter

11. Testing Status

Current Status: Tests are not implemented.

- stock/tests.py - Empty
- sale/tests.py - Empty
- No test framework configured in settings.py

Recommendation: Implement pytest or Django's test framework for comprehensive testing.

12. Summary

Mosaic Backend is a production-ready **Indian GST-compliant POS and Inventory Management Backend** built with Django REST Framework. It provides comprehensive APIs for inventory management (items, variants, batches, serial numbers), sales/POS (cart, invoicing, payments, GST calculation), purchasing (PO, GRN, purchase invoices), multi-location management (different GSTINs per warehouse), and shift management (cashier shifts with variance tracking).

All endpoints are protected by Bearer token authentication and support filtering, searching, pagination, and ordering out of the box. The system follows best practices for Indian GST compliance with automatic CGST/SGST/IGST calculation based on the origin and destination states.

End of Document